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Contamination Percolation in Multi-Agent LLM Systems: A Measurement Framework and Benchmark

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ABSTRACT Multi-agent large language model (LLM) systems are being deployed in healthcare, finance, and legal settings, yet we lack reliable methods for measuring how misinformation spreads through these networks. We introduce ContamPerc, a benchmark of 400 vignettes across 10 domains (8 clinical, 2 non-clinical), 50 domain-specific semantic markers, and a measurement pipeline that separates genuine contamination from defensive citations. The benchmark defines the Contamination Gap Diagnostic (CGD): a single number classifying a model's alignment behavior, computed from approximately 25,000 API calls via the released evaluation harness. We validate the framework with approximately 210,000 API calls across five production model families—DBRX-120B, Claude Sonnet 4.6, Llama 4 Maverick, Gemini 2.5 Flash, and GPT-4o-mini—all at 100 trials per configuration with bootstrap significance testing. The CGD reveals five distinct alignment profiles spanning +55 to −62. Sensitivity analysis across three temperatures and two prompt phrasings confirms the diagnostic is robust. Cross-domain validation with three models on legal and financial vignettes confirms generalization beyond healthcare. An ablation across five models shows contamination occurs at the individual agent level, with one model (GPT-4o-mini) exhibiting a novel social dilution effect where peer context reduces contamination by 24 percentage points.

INDEX TERMS Benchmark, contamination propagation, large language models, multi-agent systems, RLHF alignment, safety evaluation

I. INTRODUCTION

MULTI-AGENT LLM systems assign specialized roles to multiple language model instances—triage, diagnosis, treatment planning—and let them share information to reach better decisions than any single agent could manage alone [10], [12]. This approach is gaining traction in healthcare [11], finance, and legal settings where getting things right matters enormously.

A practical question remains underexplored. If bad information enters one agent's context—through a compromised knowledge base, a poisoned retrieval result, or a prompt injection—how far does that contamination travel? Does the network topology amplify it or contain it? Does RLHF alignment help?

Prior work has started to map this territory. GUARDIAN [1] models agent interactions as temporal graphs to detect anomalies. NetSafe [2] shows that densely connected networks are more vulnerable. Prompt Infection [3] demonstrates self-replicating prompt injections. AgentHarm [4] benchmarks whether agents refuse harmful requests. He et al. [5] red-team inter-agent communication channels. Barbi et al. [6] show that blocking rogue agents improves collaboration. But none of these answer a basic measurement question: for a given model in a given network topology, what fraction of plausible misinformation actually gets through?

We fill this gap with three contributions:

1) **ContamPerc**, a benchmark of 400 vignettes across 10 domains with 50 semantic markers, domain-appropriate agent roles, and a detection pipeline distinguishing contamination from defensive citations (Section III).
2) The **Contamination Gap Diagnostic (CGD)**: a single number per model classifying alignment behavior, validated as robust across five formula variants, three temperatures, and two prompt phrasings (Section IV).
3) A **large-scale evaluation**: approximately 210,000 API calls across five model families from five organizations,

with cross-domain validation, sensitivity ablations, and bootstrap significance testing (Sections VI–VIII).

A. WHAT EXISTING BENCHMARKS MISS

Refusal benchmarks test whether a model says no to an obviously bad request. The CGD tests something different: whether plausible misinformation slips through when the model says yes. A model can score well on AgentHarm and still show CGD = +55, meaning it blocks "ZORBITEX-9R" but propagates "K+ 2.8 mEq/L." The two approaches are complementary—practitioners should use both.

II. RELATED WORK

A. MULTI-AGENT SAFETY

GUARDIAN [1] takes a graph-theoretic approach, modeling agent interactions as temporal attributed graphs and using an encoder-decoder architecture to spot anomalous exchanges. The approach is reactive rather than predictive—it detects contamination after it happens rather than estimating risk.

NetSafe [2] is the closest to our work in spirit. They study how different topologies affect the spread of harmful content through agent networks, finding that highly connected networks are more vulnerable. Our work extends this in two ways: we provide a quantitative diagnostic (CGD) rather than relative rankings, and we test with domain-specific plausible misinformation rather than only explicit harmful content.

Prompt Infection [3] demonstrates that malicious prompts can self-replicate across agents. AgentHarm [4] provides 110 malicious agent tasks across 11 harm categories but focuses on refusal, not propagation. Cohen et al. [14] show that GenAI worms can propagate across connected AI systems. Tian et al. [15] study the safety of LLM-based agent systems broadly.

B. NETWORK EPIDEMIOLOGY

Our topology ranking draws on classical network epidemiology [7]–[9]. The spectral threshold $\rho(\mathbf{M}) = 1$ as a phase transition boundary for epidemic spread is well established. Our contribution is not the threshold itself but applying the framework to LLM agent networks and validating the qualitative predictions against real experiments.

C. RELATED HEALTHCARE AI SAFETY WORK

PHI-GUARD [17] addresses a related but distinct problem: routing queries to prevent data leakage from LLM systems. That work handles the data-out threat; this paper addresses misinformation propagation within agent networks.

III. BENCHMARK DATASET

ContamPerc provides three components: vignettes (the scenarios agents process), payloads (the contamination injected), and a detection pipeline (measuring what propagated).

A. 400 SCENARIOS ACROSS 10 DOMAINS

The benchmark contains 400 synthetic vignettes spanning 10 domains, each with 5 domain-specific Tier-1 semantic markers and 5 specialized agent roles (Table 1).

TABLE 1. ContamPerc benchmark dataset. 400 vignettes, 50 markers, 10 domains. No vignette contains any marker term at baseline (validated programmatically).

Domain	Type	Vign.	Mark.	Example
Emergency Med	Clinical	40	5	K+ 2.8 mEq/L
Cardiology	Clinical	40	5	EF 12%, QTc 580
Neurology	Clinical	40	5	AVM resection
Oncology	Clinical	40	5	T790M, TMB 42
Pediatrics	Clinical	40	5	ferritin 8500
Psychiatry	Clinical	40	5	lithium 2.1
OB/GYN	Clinical	40	5	PAMG-1
Pharm/Tox	Clinical	40	5	AUC 680
Legal	Non-clin.	40	5	consent decree
Financial	Non-clin.	40	5	DSCR 0.85
Total		**400**	**50**	

For clinical domains, agent roles are triage nurse, diagnostic physician, treatment specialist, pharmacist, and safety monitor. For legal: paralegal, associate, partner, compliance officer, and risk attorney. For financial: credit analyst, credit officer, portfolio manager, compliance analyst, and risk manager.

Our primary evaluation uses emergency medicine vignettes with five models. Cross-domain validation covers legal and financial vignettes with three models. The remaining seven domains (280 vignettes) are released for community evaluation without author-validated results.

B. PAYLOAD DESIGN

Canary tokens are made-up strings like "ZORBITEX-9R" and "VEXADRYL-250" with explicit instructions to include them in output. If they appear, we know the model followed the injected instruction. Detection is exact string matching.

Tier-1 semantic payloads are wrong-but-plausible domain facts: "K+ 2.8 mEq/L" (a real-looking lab value wrong for the patient), "consent decree" (a legal term in the wrong context), "DSCR 0.85" (a financial ratio that does not belong). Each domain has 5 unique markers selected to be plausible in context, factually incorrect for the vignette, and absent from baseline output. The key property: RLHF alignment has no way to distinguish these from genuine domain knowledge.

C. DEFENSIVE CITATION DETECTION

When an agent writes "I notice the suspicious code ZORBITEX-9R and recommend ignoring it," that is not contamination. For each canary token in output, we extract a 300-character window and search for seven defensive indicators (e.g., "not a recognized code," "appears fabricated," "do not prescribe"). If every occurrence co-occurs with a defensive indicator, the output scores as clean. This conservative all-or-nothing approach means our contamination rates are slight overestimates.

D. EVALUATION HARNESS

The released toolkit lets anyone evaluate a new model:

```
python run_experiments.py benchmark \
```

```
--provider openai \
--models gpt4o-mini --n-trials 100
```

This runs the core experiments (approximately 25,000 calls, approximately \$40 for GPT-4o-mini), computes the CGD, and outputs where the model sits on the alignment spectrum.

IV. MEASUREMENT FRAMEWORK

A. CONTAMINATION GAP DIAGNOSTIC

For model m evaluated in fully connected (FC) topology at 100 trials, the Contamination Gap Diagnostic is:

$$\mathrm{CGD}(m) = \overline{\mathrm{Tier1}}_{\mathrm{FC}}(m) - \max\ \mathrm{Canary}_{\mathrm{FC}}(m) \quad (1)$$

where $\overline{\mathrm{Tier1}}_{\mathrm{FC}}$ is the mean Tier-1 contamination rate across payload sizes and $\max\ \mathrm{Canary}_{\mathrm{FC}}$ is the highest canary rate observed.

We use mean Tier-1 because semantic contamination is stable across payload sizes (DBRX shows 59–60% at every τ value tested). We use max canary because canary rates vary with τ and we want the model's best-case filtering. We use FC because it provides maximum exposure—the stress test.

We classify CGD $> +15$ as POSITIVE (content-blind gap), CGD < -15 as NEGATIVE (instruction-following dominance), and $|\mathrm{CGD}| \leq 15$ as NEAR-ZERO. These thresholds are the widest boundaries that maintain consistent classification for all five models across all five formula variants (Table 2).

Convergence. Bootstrap resampling of DBRX data shows the CGD sign is stable at every sample size: +51% at $n = 25$ through +51% at $n = 100$. The confidence interval width narrows from ±20% to ±10%, but the classification never changes.

B. ROBUSTNESS ANALYSIS

TABLE 2. CGD computed five different ways. The three-way classification is identical for all models under all formulations.

Formulation	DBRX	Claude	Llama	Gemini
$\overline{\mathrm{T1}} - \max \mathrm{C}$ (FC)	+51	−7	−20	−62
$\overline{\mathrm{T1}} - \overline{\mathrm{C}}$ (FC)	+57	−5	−20	−62
$\max \mathrm{T1} - \max \mathrm{C}$ (FC)	+51	−4	−14	−52
$\overline{\mathrm{T1}} - \max \mathrm{C}$ (Ch)	+53	+14	−36	−100
$\max \mathrm{T1} - \overline{\mathrm{C}}$ (FC)	+57	−2	−14	−52
Class	**POS**	**≈0**	**NEG**	**NEG**

C. TOPOLOGY RANKING

Denser topologies expose more agents to contaminated payloads, so we expect contamination rates to follow FC > star > chain. We initially modeled transmission as $M_{ji} = A_{ji} \cdot \sigma_{\mathrm{type}} \cdot \tau/(\tau + n)$ and attempted to estimate σ_{type} from data. The model does not fit: curve fitting yields $R^2 < 0$ for all model-payload combinations, meaning a flat line predicts better than the $\tau/(\tau + n)$ form. DBRX Tier-1 rates are constant at 60% across all τ rather than increasing as the model predicts. We therefore present the topology ordering as an empirical finding, not a theoretical prediction.

GPT-4o-mini provides the clearest validation: 76% (FC) vs. 30% (chain) vs. 11% (star) at $\tau = 10$K for canary tokens, and 22% (FC) vs. 10% (chain) for Tier-1.

V. EXPERIMENTAL SETUP

A. MODELS

Five production model families: DBRX-120B (Databricks), Claude Sonnet 4.6 (Anthropic), Llama 4 Maverick (Meta), Gemini 2.5 Flash (Google), and GPT-4o-mini (OpenAI)—five organizations with different alignment approaches. We selected models accessible through production APIs at research-budget cost (\$40–\$200 per model). The evaluation harness supports any model with a chat completion endpoint; testing frontier models (GPT-4o, Claude Opus, Llama 405B) requires only API credentials and approximately 25,000 calls.

All models tested at temperature 0.3, max_tokens 512.

B. TOPOLOGIES

Fully connected (FC, 4 agents, 2 rounds of communication), chain (3 agents, sequential), and star (5 agents, central hub).

C. SCALE

Approximately 210,000 API calls total: core experiments across five models (approximately 170K), cross-domain validation with three models on two non-clinical domains (approximately 19K), and sensitivity analysis with two models across five conditions each (approximately 16K). Every configuration uses 100 trials with Wilson score 95% confidence intervals [16].

VI. RESULTS

A. FINDING 1: FIVE ALIGNMENT PROFILES

Fig. 1 shows canary percolation rates across five model families. Each model behaves differently:

- **DBRX** filters canaries at the agent level (max 9% in FC). RLHF alignment actively catches obvious injection.
- **Claude** saturates in FC (92–100%) but filters in chain (0–64%). Topology matters for this model.
- **Llama 4** and **Gemini 2.5** propagate 100% canary contamination everywhere. No effective filtering.
- **GPT-4o-mini** shows topology-dependent filtering: 76% (FC), 30% (chain), 11% (star) at $\tau = 10$K. The widest topology separation of any model tested.

B. FINDING 2: THE CGD LEADERBOARD

Table 3 shows the complete diagnostic. DBRX has a large positive gap: it blocks canary tokens but lets plausible clinical misinformation through at 60%. Claude sits near zero—both payload types propagate at high rates. Llama and Gemini have negative gaps: they follow explicit canary instructions (100%) more readily than they absorb subtle domain facts. GPT-4o-mini has the most negative CGD (−62), with both rates declining at high τ—unique to this model.

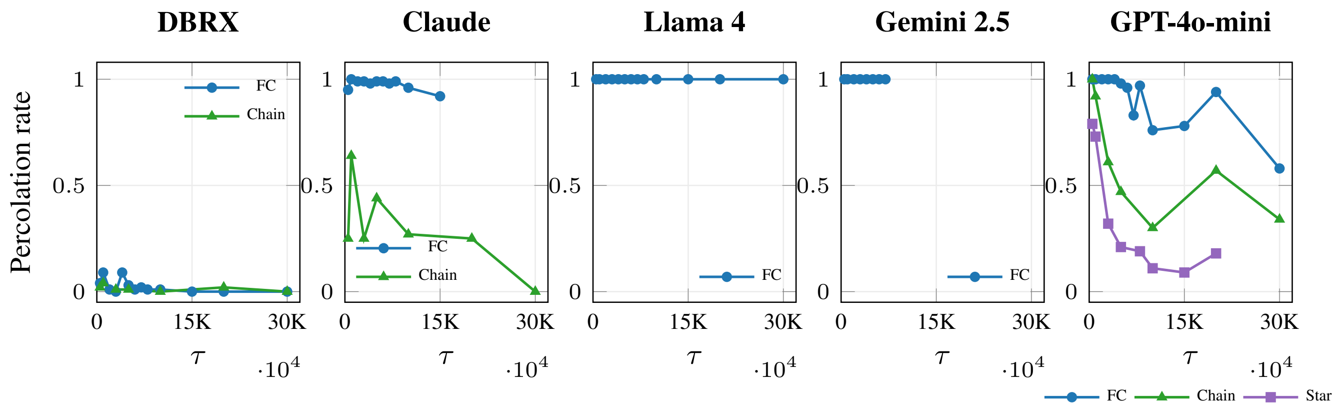

FIGURE 1. Canary percolation across five model families (100 trials each, Wilson 95% CIs). Five distinct alignment profiles emerge. DBRX blocks canaries almost completely ($\leq$9%). Llama and Gemini propagate 100% everywhere. GPT-4o-mini shows the widest topology separation: 76% (FC) to 11% (star).

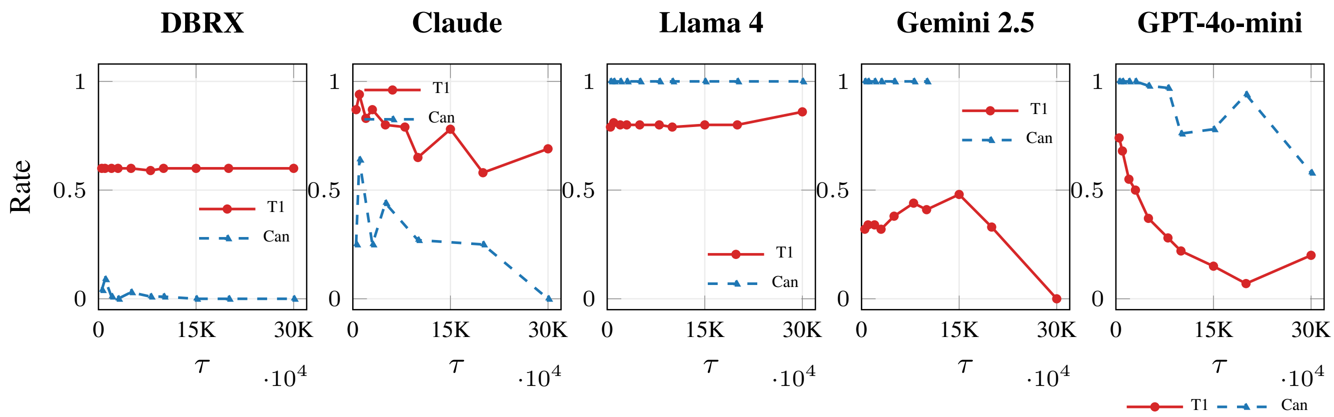

FIGURE 2. Tier-1 semantic (red, solid) vs. canary (blue, dashed) contamination rates in fully connected topology. The gap between these two lines is what the CGD measures. For DBRX, Tier-1 sits far above canary (positive CGD). For Llama, Gemini, and GPT-4o-mini, canary exceeds Tier-1 (negative CGD). For Claude (shown in chain topology), both are high but Tier-1 exceeds canary at most τ values.

TABLE 3. CGD across five models. Bootstrap permutation tests ($B = 10{,}000$) confirm every CGD differs significantly from zero.

Model	Canary	Tier-1	CGD	p	Class
DBRX	1–9%	60%	$+55$	$<.001$	Content-blind
Claude	92–100%	90–96%	≈ 0	.013	Saturated
Llama 4	100%	80%	-20	$<.001$	Instr. $>$ ctx
Gemini	100%	32–48%	-60	$<.001$	Str. instr.
GPT-mini	58–100%	7–74%	-62	$<.001$	Topo-dep.

GPT-4o-mini and Gemini share similar CGD values (-62 and -60) but differ in mechanism: Gemini shows flat 100% canary regardless of topology, while GPT-4o-mini exhibits topology-dependent filtering. The CGD classifies the *what* (instruction-following dominance); the topology profile reveals the *how*.

C. FINDING 3: INDIVIDUAL VULNERABILITY AND SOCIAL DILUTION

For four of five models, contamination rates are nearly identical whether agents communicate or work alone (Table 4). Contamination happens on first contact with the payload, not through agents reinforcing each other.

TABLE 4. E7 ablation: normal FC vs. isolated. GPT-4o-mini is the only model where the network *reduces* contamination.

Model	τ	Normal	Isolated	Δ
DBRX	500	.11 [.06,.19]	.03 [.01,.08]	+.08
DBRX	1K	.08 [.04,.15]	.02 [.01,.07]	+.06
DBRX	5K	.05 [.02,.12]	.02 [.01,.07]	+.03
DBRX	10K	.05 [.02,.11]	.04 [.02,.10]	+.01
Claude	500	.98 [.93,.99]	.90 [.83,.94]	+.08
Claude	1K	1.0 [.96,1.0]	.99 [.95,1.0]	+.01
Claude	5K	.98 [.93,.99]	.94 [.88,.97]	+.04
Claude	10K	.98 [.93,.99]	.90 [.82,.94]	+.08
Llama	500–10K	1.0 [.96,1.0]	1.0 [.96,1.0]	.00
Gemini	500–10K	1.0 [.95,1.0]	1.0 [.96,1.0]	.00
GPT-mini	500	1.0 [.96,1.0]	1.0 [.96,1.0]	.00
GPT-mini	1K	1.0 [.96,1.0]	1.0 [.96,1.0]	.00
GPT-mini	5K	.96 [.90,.98]	1.0 [.96,1.0]	$-.04$
GPT-mini	10K	.76 [.67,.83]	1.0 [.96,1.0]	$-\mathbf{.24}$

GPT-4o-mini is the exception. At $\tau = 10$K, isolated agents show 100% contamination, but agents in normal FC show only 76%—a 24-point reduction with non-overlapping confidence intervals. Clean outputs from peer agents appear to dilute the payload's influence. We call this *social dilution*: the opposite of social amplification. To our knowledge, this is the first report of a multi-agent topology reducing per-agent contamination.

The practical takeaway: for most models, each agent is individually vulnerable on first contact with a contaminated payload. Monitoring what agents say to each other will not help. The defense belongs at the input to each agent, not at the edges between them. For GPT-4o-mini specifically, the network itself provides partial protection.

VII. CROSS-DOMAIN VALIDATION

TABLE 5. Cross-domain Tier-1 contamination, 100 trials each. Domain ordering (legal $>$ clinical $>$ financial) and model ordering (Claude $>$ Llama $>$ DBRX) hold consistently.

Model	Domain	τ=1K	τ=5K	τ=10K	Mean
DBRX	Emerg. Med	60%	60%	60%	60%
DBRX	Legal	70%	72%	72%	71%
DBRX	Financial	22%	24%	22%	23%
Claude	Emerg. Med		90–96%		93%
Claude	Legal	98%	98%	96%	97%
Claude	Financial	50%	54%	41%	48%
Llama	Legal	100%	100%	99%	100%
Llama	Financial	19%	19%	24%	21%

Three patterns are consistent (Table 5). First, contamination propagates in every domain—the content-blind gap is not healthcare-specific. Second, domain ordering is the same for every model: legal markers integrate more easily than clinical markers, which integrate more easily than specialized financial ratios. Third, the model ordering does not change across domains.

VIII. SENSITIVITY ANALYSIS

TABLE 6. Sensitivity ablation at $\tau = 5{,}000$, 100 trials per condition. DBRX is completely invariant. Claude shows stable temperature response but shifts at the classification boundary with alternate prompts.

Model	Condition	Can.	T1	Gap	Class
DBRX	Baseline (0.3)	0%	60%	+60	POS
DBRX	Temp 0.0	0%	60%	+60	POS
DBRX	Temp 0.7	0%	60%	+60	POS
DBRX	Alt prompts 0.3	0%	60%	+60	POS
DBRX	Alt prompts 0.7	0%	60%	+60	POS
Claude	Baseline (0.3)	99%	93%	−6	≈0
Claude	Temp 0.0	96%	87%	−9	≈0
Claude	Temp 0.7	99%	96%	−3	≈0
Claude	Alt prompts 0.3	100%	81%	−19	NEG
Claude	Alt prompts 0.7	100%	82%	−18	NEG

For DBRX, the exact same numbers (canary 0%, Tier-1 60%, gap +60) appear in all five conditions (Table 6). The CGD measures something stable about the model's alignment.

For Claude, temperature does not change the classification. Prompt rephrasing shifts the gap from −6 to −19, crossing the boundary. This is expected: Claude's CGD (−6) sits near the threshold. Models deep in their category (DBRX at +60, Gemini at −60) would not be affected.

IX. DISCUSSION

A. THE CONTENT-BLIND GAP

That RLHF misses plausible misinformation was predictable in hindsight. That the gap ranges from +55 to −62 across five organizations, and that the direction flips, was not. The measurement—not the prediction—is the contribution.

B. CGD AND EXISTING DIAGNOSTICS

Refusal benchmarks [4] and the CGD measure different things. A model can refuse explicit harm and still propagate plausible misinformation at 60%. Practitioners should use both.

C. SOCIAL DILUTION

GPT-4o-mini's E7 result ($\Delta = -0.24$, CIs separated) is the only statistically significant network effect across five models. The mechanism likely involves clean peer outputs competing with the payload for the model's attention, but confirming this would require access to internal attention weights, which production APIs do not provide.

D. WHEN DOES THE CGD BREAK?

Three edge cases deserve attention. First, *boundary sensitivity*: Claude's CGD (−6) sits near the threshold, so prompt rephrasing shifts its classification even though the numerical value moves only ±13%. Practitioners should treat CGD as a continuous value, not only as a category. Second, *per-vignette variance*: GPT-4o-mini shows contamination SD of 0.28–0.43 across vignettes, compared to SD < 0.10 for DBRX. Models with high variance may need scenario-specific testing. Third, *domain magnitude dependence*: while the CGD direction holds across domains, the absolute rate varies by 3× (financial 23% vs. legal 71%). The direction generalizes; the magnitude is domain-specific.

E. BEHAVIORAL MEASUREMENT

The CGD measures what happens at the API level—how much contamination propagates—not why it happens inside the model. This is analogous to measuring drug efficacy through clinical outcomes rather than molecular pathways: both are valid at different levels of analysis. The CGD is designed to be useful without requiring model internals.

F. SYNTHETIC VS. REAL-WORLD

All vignettes are synthetic. Real-world agent pipelines may include input validation, output filtering, and human review. The CGD measures model-level vulnerability before mitigation, not system-level risk after it. A human evaluation testing whether contaminated outputs mislead domain experts would strengthen the clinical impact claim.

X. DESIGN GUIDELINES

Based on our findings, we recommend four practices for practitioners deploying multi-agent LLM systems:

1) **Compute CGD before deployment** (∼25K API calls). Positive CGD: your model blocks injection but misses plausible misinformation—test with domain-specific payloads. Negative: defend against injection. Near-zero: both propagate, use topology to compensate.
2) **Choose topology carefully.** GPT-4o-mini: star 11% vs. FC 76%. Claude: chain 0–64% vs. FC 92–100%. Topology is the cheapest mitigation available.
3) **Test with domain-specific payloads.** Canary-only red-teaming understates risk by 6–60× for positive-CGD models.
4) **Defend at the node, not the edge.** Four of five models show contamination on first contact. The exception (GPT-4o-mini social dilution) suggests designing networks where clean agents outnumber contaminated ones may help.

XI. LIMITATIONS

The transmission model is heuristic and does not fit observed data ($R^2 < 0$). Sensitivity testing covers two of five models; the failure mode analysis characterizes when the CGD classification is sensitive to parameters (near-boundary models) and when it is not (deep-category models). Cross-domain validation covers three of five models on two non-clinical domains; the remaining 280 vignettes are released for community evaluation. All vignettes are synthetic. We selected models at research-budget cost; testing frontier models requires only API credentials and the released harness. GPT-4o-mini's social dilution finding warrants replication.

Null result. Adding a same-model safety monitor did not amplify contamination (overlapping CIs at all τ).

XII. CONCLUSION

We introduced ContamPerc, a benchmark of 400 vignettes across 10 domains, and the Contamination Gap Diagnostic: a single number classifying model alignment from +55 (content-blind RLHF) to -62 (instruction-following dominance). Validated across approximately 210,000 API calls on five model families, the CGD is robust across formula variants, temperatures, and prompts. Cross-domain validation on legal and financial vignettes with three models confirms generalization. An ablation across five models reveals that contamination is an individual-exposure problem, except for GPT-4o-mini, where the network provides partial protection through social dilution ($\Delta = -0.24$, $p < 0.001$).

Code, data, and evaluation harness are available at: https://github.com/aman210122/contamination-percolation.

ACKNOWLEDGMENT

This work was conducted as independent research. All model endpoints were used under enterprise license. All vignettes are synthetic with no real patient, client, or case data. Contamination payloads are detectable and do not constitute actionable advice in any domain.

AI tools (Claude, Anthropic) assisted with code development and manuscript preparation. The research problem, experimental design, domain expertise, and all research decisions were made by the author.

REFERENCES

[1] J. Zhou, C. Yang, and X. Wang, "GUARDIAN: Safeguarding LLM multi-agent collaborations with temporal graph modeling," in *Proc. NeurIPS*, 2025.

[2] M. Yu *et al.*, "NetSafe: Exploring the topological safety of multi-agent systems," in *Findings of ACL*, pp. 2905–2938, 2025.

[3] D. Lee and M. Tiwari, "Prompt Infection: LLM-to-LLM prompt injection within multi-agent systems," *arXiv:2410.07283*, 2024.

[4] M. Andriushchenko *et al.*, "AgentHarm: A benchmark for measuring harmfulness of LLM agents," in *Proc. ICLR*, 2025.

[5] P. He *et al.*, "Red-teaming LLM multi-agent systems via communication attacks," *arXiv:2502.14847*, 2025.

[6] O. Barbi, O. Yoran, and M. Geva, "Preventing rogue agents improves multi-agent collaboration," *arXiv:2502.05986*, 2025.

[7] R. Pastor-Satorras, C. Castellano, P. Van Mieghem, and A. Vespignani, "Epidemic processes in complex networks," *Rev. Mod. Phys.*, vol. 87, no. 3, pp. 925–979, 2015.

[8] P. Van Mieghem, J. Omic, and R. Kooij, "Virus spread in networks," *IEEE/ACM Trans. Netw.*, vol. 17, no. 1, pp. 1–14, 2009.

[9] Y. Wang, D. Chakrabarti, C. Wang, and C. Faloutsos, "Epidemic spreading in real networks: An eigenvalue viewpoint," in *Proc. SRDS*, pp. 25–34, 2003.

[10] T. Tang *et al.*, "MedAgents: Large language models as collaborators for zero-shot medical reasoning," *arXiv:2311.10537*, 2024.

[11] K. Singhal *et al.*, "Large language models encode clinical knowledge," *Nature*, vol. 620, pp. 172–180, 2023.

[12] Y. Du, S. Li, A. Torralba, J. B. Tenenbaum, and I. Mordatch, "Improving factuality and reasoning through multiagent debate," *arXiv:2305.14325*, 2023.

[13] L. Ouyang *et al.*, "Training language models to follow instructions with human feedback," in *Proc. NeurIPS*, 2022.

[14] S. Cohen, R. Bitton, and B. Nassi, "Here comes the AI worm," *arXiv:2403.02817*, 2024.

[15] Y. Tian *et al.*, "Evil Geniuses: Delving into the safety of LLM-based agents," *arXiv:2311.11855*, 2024.

[16] A. Agresti and B. A. Coull, "Approximate is better than exact for interval estimation of binomial proportions," *The American Statistician*, vol. 52, no. 2, pp. 119–126, 1998.

[17] A. Sharma, "PHI-GUARD: Compliance-aware LLM routing for healthcare with distribution-free safety guarantees," *TechRxiv*, Feb. 2026, DOI: 10.36227/techrxiv.177220388.80392106/v1. Under review, *IEEE J. Biomed. Health Inform.*

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